

COMP1017 (Mathematics for Computer Scientists) Test 1

Academic Session 2018-2019, Autumn Semester

This is an open book test. There are THREE (3) questions pertaining to propositional logic, predicate logic, sets, and functions. Answer ALL the questions. Time allowed 60 minutes.

1. Find a compound proposition involving the propositional variables p , q , and r that is true when p and q are true and r is false, but is false otherwise.

[10 marks]

2. Personal data means any information in respect of commercial transaction in which a data subject can be identified, or is identifiable with other information.

(a) Identify atomic predicates in the above sentence.

[20 marks]

(b) Use quantifiers to express the above sentence with the atomic predicates you identified in (a). Show your work.

[20 marks]

[Note: there is no standard answer to question 2. Marks are awarded as long as the answers to (a) and (b) are consistent and logically correct.]

3. Let $S = \{1, 2, \dots, 10\}$ be a set of first 10 positive integers. Let $A, B, C \subseteq S$. Define the operator Δ such that $A \Delta B = \overline{A \cap B} - \overline{A \cup B}$, where the universal set is S . Let $f(A, B) = (\{b | \forall a \in A, \exists b \in B, a \geq b\}, \{a | \exists a \in A, \forall b \in B, a \geq b\})$.

(a) Draw a Venn diagram for $A \Delta B$ and show that $A \Delta B = (A \cup B) - (A \cap B)$.

[10 marks]

(b) For the operator Δ , specify its domain, the cardinality of the domain, codomain and the cardinality of codomain.

[15 marks]

(c) Determine whether the composition $f \circ \Delta$ and $\Delta \circ f$ exist and if they do find $f \circ \Delta (A, B)$ and $\Delta \circ f(A, B)$ for $A = \{2, 4, 5, 6\}$ and $B = \{1, 2, 3, 4, 6, 7\}$.

[25 marks]